CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 19: Constructions (Circles)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Circumscribed circle of a triangle

1. Using ruler and compasses only, construct $\triangle ABC$ with $AB = 6$ cm, $AC = 4.5$ cm and $\angle BAC = 120^\circ$. Construct the circle that passes through A, B and C, and measure its radius. ICSE 2025, 4 marks

SOLUTION

Draw $AB = 6$ cm. At A construct an angle of 120° (two 60° arcs) and cut off $AC = 4.5$ cm on its arm. Join BC.

Draw the perpendicular bisectors of AB and AC. Let them meet at O.

With centre O and radius OA, draw the circle. It passes through A, B and C.

Because $\angle A$ is obtuse, O lies outside the triangle, on the other side of BC from A.

Check by calculation: $BC^2 = 6^2 + 4.5^2 - 2(6)(4.5)\cos 120^\circ = 83.25$, so $BC \approx 9.12$ cm and the radius is $\dfrac{BC}{2\sin 120^\circ} \approx 5.27$ cm.

Answer: the radius measures about 5.3 cm

2. Using ruler and compasses, construct $\triangle ABC$ with $AB = 6$ cm, $\angle BAC = 120^\circ$ and $AC = 5$ cm. Construct the circle passing through A, B and C, and measure its radius. ICSE 2023, 4 marks

SOLUTION

Draw $AB = 6$ cm. At A construct an angle of 120° and cut off $AC = 5$ cm. Join BC.

Draw the perpendicular bisectors of AB and AC, meeting at O. With centre O and radius OA, draw the circle.

Check by calculation: $BC^2 = 6^2 + 5^2 - 2(6)(5)\cos 120^\circ = 91$, so $BC \approx 9.54$ cm and the radius is $\dfrac{BC}{2\sin 120^\circ} \approx 5.5$ cm.

Answer: the radius measures about 5.5 cm

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3. Using ruler and compasses, construct $\triangle ABC$ with $AB = 3$ cm, $BC = 4$ cm and $\angle ABC = 90°$. Construct its circumscribing circle and measure the radius. ICSE 2020, 4 marks

SOLUTION

Draw $BC = 4$ cm, construct a right angle at B and cut off $BA = 3$ cm. Join AC.

Draw the perpendicular bisectors of AB and BC, meeting at O. With centre O and radius OA, draw the circle.

Check: in a right-angled triangle the circumcentre is the mid-point of the hypotenuse. $AC = \sqrt{3^2 + 4^2} = 5$ cm, so the radius is 2.5 cm.

Answer: radius $= 2.5$ cm

4. Using ruler and compasses only, construct $\triangle ABC$ with $BC = 5$ cm, $AB = 6.5$ cm and $\angle ABC = 120°$. (a) Construct its circumcircle. (b) Construct the cyclic quadrilateral $ABCD$ in which D is equidistant from AB and BC. ICSE 2018, 4 marks

SOLUTION

Draw $BC = 5$ cm. At B construct an angle of $120°$ and cut off $BA = 6.5$ cm. Join AC.

(a) Draw the perpendicular bisectors of AB and BC, meeting at O. With centre O and radius OB, draw the circle through A, B and C.

(b) Points equidistant from AB and BC lie on the bisector of $\angle ABC$. Draw this bisector and let it meet the circle again at D. Join AD and CD.

Check: the bisector cuts the arc AC in half, so $AD = CD$, and $\angle ADC = 180° - 120° = 60°$. So $\triangle ADC$ should be equilateral.

Answer: $ABCD$ is the required cyclic quadrilateral, with $AD = CD$

2 Tangents from an external point

5. Using ruler and compasses, draw a circle of radius 4 cm with centre O. Mark a point P 7 cm from O and construct the two tangents from P to the circle. Measure the length of a tangent. ICSE 2019, 4 marks

SOLUTION

Draw the circle and mark P with $OP = 7$ cm.

Bisect OP at M. With centre M and radius MO, draw a circle cutting the given circle at A and B.

Join PA and PB. These are the tangents, because $\angle OAP = 90°$ (angle in a semicircle).

Check: $PA = \sqrt{7^2 - 4^2} = \sqrt{33} \approx 5.74$ cm.

Answer: each tangent measures about 5.7 cm

3 Inscribed circle of a triangle

6. Using ruler and compasses, construct $\triangle ABC$ with $AB = 7$ cm, $\angle CAB = 60^\circ$ and $AC = 5$ cm. Construct the locus of points equidistant from (a) AB and AC (b) BA and BC. Hence construct the circle that touches the three sides of the triangle internally. ICSE 2017, 4 marks

SOLUTION

Draw $AB = 7$ cm, construct $\angle CAB = 60^\circ$ and cut off $AC = 5$ cm. Join BC.

(a) The locus is the bisector of $\angle A$. (b) The locus is the bisector of $\angle B$.

The bisectors meet at I, the incentre. From I draw $IN \perp AB$. With centre I and radius IN, draw the circle. It touches all three sides.

Check: $BC = \sqrt{49 + 25 - 35} = \sqrt{39} \approx 6.24$ cm, the area is $\frac{1}{2} \times 7 \times 5 \times \sin 60^\circ \approx 15.16$ cm² and the semi-perimeter is about 9.12 cm, so the radius is about $\frac{15.16}{9.12} \approx 1.7$ cm.

Answer: the incircle has a radius of about 1.7 cm

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